

# karAQI: A Serverless End-to-End Machine Learning Pipeline for 72-Hour Air Quality Index Forecasting in Karak, Pakistan

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## Abstract

This paper presents karAQI, an end-to-end machine learning system for forecasting the US EPA Air Quality Index (AQI) up to 72 hours in advance for Karak, Pakistan—a city with no local ground-air-quality monitor. The system fetches hourly pollutant and weather data from Open-Meteo (a free, keyless API wrapping Copernicus CAMS reanalysis data), engineers 33 hourly features, and trains two production models—Ridge regression and XGBoost—selected from an initial pool of six model families (Ridge, Random Forest, XGBoost, SARIMA, LSTM, and persistence baselines). The production model produces 30 predictions per forecast origin: 24 hourly point estimates, four six-hour block means, and two twelve-hour block means spanning a 72-hour horizon. The system operates on a serverless, zero-cost stack: a DuckDB feature store, MLflow file-backed model registry, and GitHub Actions CI/CD pipelines (hourly feature collection, hourly forecast generation, daily model retraining). Predictions are pre-computed by CI and served as static JSON files via a Streamlit dashboard with near-instant page load. Rolling-origin evaluation with a 72-hour embargo shows the production XGBoost model achieves RMSE of 8.89 on hourly points, consistently outperforming persistence and seasonal persistence baselines on all output groups. The system is fully automated, publicly deployed, and designed for reproducibility.

**Keywords:** Air Quality Index, AQI Forecasting, Machine Learning, XGBoost, Ridge Regression, Time Series, MLOps, Serverless, Open-Meteo, Streamlit, CI/CD

## 1 Introduction

Air quality is a critical public health concern, particularly in South Asian cities where industrial activity, vehicular emissions, dust storms, and seasonal biomass burning contribute to hazardous pollution levels. The World Health Organization estimates that air pollution causes 7 million premature deaths annually worldwide [1]. Timely AQI forecasts enable citizens to plan outdoor activities, healthcare systems to prepare for pollution-related admissions, and policymakers to issue timely advisories.

Karak, a city in Khyber Pakhtunkhwa, Pakistan (33.14°N, 71.19°E), presents a unique challenge: it has no local ground-air-quality monitoring station. This means that any AQI forecasting system must rely entirely on modeled or reanalysis data from remote sensing and atmospheric modeling services, rather than direct ground-truth measurements. This constraint shapes every design decision in the system presented here.

This paper presents **karAQI**, a complete machine learning pipeline that:

1. Fetches hourly pollutant (PM<sub>2.5</sub>, PM<sub>10</sub>, CO, NO<sub>2</sub>, SO<sub>2</sub>, O<sub>3</sub>) and weather data from the Open-Meteo API (free, keyless, powered by Copernicus CAMS reanalysis).

2. Engineers 33 hourly features including temporal encoding, lagged values, and rolling statistics.
3. Trains and evaluates six model families (Ridge, Random Forest, XGBoost, SARIMA, LSTM, persistence baselines) using rolling-origin cross-validation with a 72-hour embargo, selecting the two best for production.
4. Serves pre-computed 72-hour forecasts via a Streamlit dashboard with near-instant page load.
5. Automates the entire pipeline with three GitHub Actions workflows (hourly feature collection, hourly forecast generation, daily model retraining).
6. Operates on a serverless, zero-cost infrastructure with no API keys and no paid services.

The paper is organized as follows: Section 2 reviews related work. Section 3 describes data collection and preprocessing. Section 4 details feature engineering, model selection, and training. Section 5 presents the system architecture. Section 6 reports results and discussion. Section 7 concludes.

## 2 Related Work

AQI forecasting has been extensively studied using various approaches:

**Statistical models.** ARIMA and SARIMA models capture seasonal patterns in air quality time series. Comparative analyses [2] identify SARIMA as a strong baseline for AQI prediction, particularly when seasonal decomposition reveals clear periodic patterns.

**Machine learning models.** Random Forest, XGBoost, and gradient boosting methods have shown strong performance on tabular air quality data. Research on ML models for daily AQI prediction [3] finds Extra Trees and Random Forest most effective, with  $PM_{2.5}$ ,  $PM_{10}$ , and  $O_3$  as the most important features. Linear models such as Multiple Linear Regression (MLR) and Ridge regression serve as interpretable baselines [4].

**Deep learning models.** LSTM networks are widely used for air quality forecasting due to their ability to capture long-range temporal dependencies [5]. However, their effectiveness depends on dataset size; with limited training data, simpler models often outperform deep networks due to lower variance.

**Satellite and reanalysis data.** Studies using satellite-derived data (Sentinel-5P TROPOMI) demonstrate that modeled/gridded atmospheric data can represent AQI patterns in regions without ground monitors [6]. This is directly relevant to Karak, where Copernicus CAMS reanalysis data from Open-Meteo serves as the sole data source.

**MLOps and deployment.** While most AQI research focuses on model accuracy, fewer studies address the full MLOps lifecycle: feature stores, model registries, CI/CD automation, and production deployment. This paper contributes a complete, reproducible MLOps pipeline.

## 3 Data Collection and Preprocessing

### 3.1 Data Sources

The system uses a single data provider—Open-Meteo—for all inputs. Open-Meteo wraps Copernicus CAMS (air quality) and ERA5/IFS (weather) reanalysis data, providing free, keyless access to hourly atmospheric data at any global coordinate.

Two Open-Meteo endpoints are used:

- **Air Quality API** (`air-quality-api.open-meteo.com`):  $PM_{2.5}$ ,  $PM_{10}$ , CO, NO<sub>2</sub>, SO<sub>2</sub>, O<sub>3</sub>, aerosol optical depth, dust, UV index. Historical coverage from 2021 onward.
- **Weather Archive API** (`archive-api.open-meteo.com`): temperature, humidity, dew point, precipitation, rain, pressure, cloud cover, wind speed/direction/gusts. Historical coverage from 2000 onward.

A third endpoint—the Open-Meteo AQ Forecast API—provides a 96-hour (4-day) US AQI reference line for dashboard comparison, trimmed to the 72-hour window from the forecast origin. This uses the same EPA AQI scale as the training target.

### 3.2 Data Quality and Sanity Checks

Before committing to Open-Meteo as the sole provider, a cross-source validation was performed against OpenWeather and WAQI/AQICN:

- **Timezone normalization** was the dominant source of apparent disagreement. Without normalization,  $\text{PM}_{2.5}$  correlation between Open-Meteo and OpenWeather was  $\sim 0.4$ . After aligning both sources to Asia/Karachi (UTC+5), correlation improved to 0.68 (hourly) and 0.77 (daily).
- **WAQI/Peshawar station** (nearest to Karak,  $\sim 120$  km) reported data from March 2025—over 4 months stale at the time of checking. Not usable as a live reference.
- **IQAir rate-limiting** blocked bulk historical pulls, making it unsuitable for training data collection.
- **Remaining cross-source differences** (correlation 0.57–0.70 across pollutants) were attributed to expected model-to-model disagreement for a rural area with complex topography, not to data quality issues.

### 3.3 Historical Data Gap

The first Open-Meteo AQ pull (2022-08-01 to present) revealed a contiguous upstream null block from 2022-08-01 00:00 through 2022-08-04 04:00 (77 timestamps, all nine AQ variables null). Weather values existed for these timestamps. Rather than imputing unavailable model outputs, the active training range begins from 2022-08-05.

### 3.4 Incremental Fetching

The feature pipeline runs hourly and fetches only new data since the last pull, rather than re-downloading the full 4-year history on every run. This is possible because Open-Meteo reanalysis data is immutable—historical values never change. At training time, typically only  $\sim 1$  hour of new data is fetched instead of  $\sim 1,460$  days. The first run performs a full fetch; every subsequent run takes under 1 second instead of  $\sim 5$  seconds.

### 3.5 Dataset Summary

Table 1: Dataset characteristics

| Property             | Value                | Notes                                        |
|----------------------|----------------------|----------------------------------------------|
| Location             | 33.14°N, 71.19°E     | Karak, Sabir Abad area                       |
| Timezone             | Asia/Karachi (UTC+5) | Explicit in all API requests                 |
| AQI training range   | 2022-08-05 – present | After 77-hour upstream gap                   |
| Weather trend range  | 2000-01-01 – present | Daily summaries for long-run analysis        |
| Hourly training rows | $\sim 35,400$        | 4 years of hourly data                       |
| Daily training rows  | $\sim 1,470$         | 4 years of daily aggregates                  |
| AQI standard         | US EPA               | May 2024 $\text{PM}_{2.5}$ breakpoint update |

## 4 Methodology

### 4.1 AQI Target Calculation

The target variable is computed using the official US EPA AQI breakpoint method implemented in `src/aqi.py`. For each hour:

1. Each pollutant’s raw concentration is converted to a sub-index using pollutant-specific averaging windows (24h for  $\text{PM}_{2.5}$ , 8h for  $\text{O}_3/\text{CO}$ , 1h for  $\text{SO}_2/\text{NO}_2$ ), unit conversions, and EPA breakpoint interpolation.

2. The hourly AQI is the **maximum valid pollutant sub-index**.
3. Categories are assigned: Good (0–50), Moderate (51–100), Unhealthy for Sensitive Groups (101–150), Unhealthy (151–200), Very Unhealthy (201–300), Hazardous (301+).

This is a modeled AQI estimate derived from CAMS reanalysis concentrations, not a station-measured value.

## 4.2 Feature Engineering

The hourly model uses 33 features per timestamp:

Table 2: Feature categories

| Category         | Features                                                                                                      | Count     |
|------------------|---------------------------------------------------------------------------------------------------------------|-----------|
| Pollutant inputs | PM <sub>2.5</sub> , PM <sub>10</sub> , CO, NO <sub>2</sub> , SO <sub>2</sub> , O <sub>3</sub> , AOD, dust, UV | 10        |
| Weather inputs   | temp, humidity, dew point, precip, rain, pressure, cloud, wind (3)                                            | 10        |
| Calendar         | hour sin/cos, DOW sin/cos, month sin/cos, is_weekend                                                          | 7         |
| Lag features     | AQI lag 1h/24h                                                                                                | 2         |
| Rolling features | AQI rolling 6h/24h mean and std                                                                               | 4         |
| <b>Total</b>     |                                                                                                               | <b>33</b> |

The daily model uses 39 features with similar categories adapted to daily resolution (e.g., PM<sub>2.5</sub> lag 1/2/3/7 days, AQI rolling 3/7 day mean and std).

All features are historical-only—no future weather or pollutant values are used as inputs. The forecast origin is the latest completed hourly row.

## 4.3 Model Selection

Model selection followed a structured methodology rather than ad-hoc experimentation. Five research papers were reviewed, and nine candidate models were scored on seven weighted criteria:

Table 3: Model selection scoring criteria

| Criterion                 | Weight | Description                                         |
|---------------------------|--------|-----------------------------------------------------|
| Published accuracy        | 0.20   | High reported accuracy/R <sup>2</sup> across papers |
| Stability / low variance  | 0.15   | Consistent results, low overfitting risk            |
| Relevance to modeled data | 0.15   | Works with CAMS/reanalysis inputs                   |
| Small-sample fitness      | 0.15   | Performs on short series ( $n \approx 1,457$ daily) |
| Interpretability          | 0.10   | SHAP/LIME support                                   |
| Assignment stack fit      | 0.15   | Matches required models (RF, Ridge, TF)             |
| Deployment feasibility    | 0.10   | Free, no API keys, runs on student hardware         |

A stratified selection rule ensured coverage across model families: highest-scoring model from each family (boosting → XGBoost, bagging → Random Forest, statistical → SARIMA, deep learning → LSTM). Ridge regression was included as a mandated baseline.

After initial evaluation, **two models were selected for production**: Ridge regression (fast, interpretable, stable across all horizons) and XGBoost (best RMSE on hourly points, captures non-linear interactions). Random Forest, SARIMA, and LSTM were retained as development-only comparison models—Random Forest for deployment simplicity, LSTM for 1+ hour CPU training time with the worst RMSE, and SARIMA for negative R<sup>2</sup> across all horizons.

## 4.4 Training Protocol

**Daily models.** Each horizon (+1, +2, +3 days) is trained independently with chronological 80/20 holdout and a 3-day purge gap. Hyperparameter tuning uses `TimeSeriesSplit(gap=3)` on training data only.

**Hourly models.** Two models are trained for the 30-output hourly forecast: Ridge (multi-output, alpha selected from purged `TimeSeriesSplit`) and XGBoost (`MultiOutputRegressor`). The training frame has ~35,400 hourly rows with a 72-hour purge gap. The best model is selected by lowest average RMSE across all three output groups. A champion comparison ensures only models that improve on the current best are promoted.

## 4.5 Evaluation Protocol

Instead of a single train/test split, the project uses **expanding rolling-origin evaluation**:

1. Train on data up to day  $N$ .
2. Test on days  $N + 1$  to  $N + 3$  (72 hours).
3. Roll forward: train on data up to day  $N + 7$ .
4. Test on days  $N + 8$  to  $N + 10$ .
5. Repeat 3 times.

The 72-hour embargo between training and test sets prevents data leakage (today's lag features do not contain tomorrow's values). Metrics reported per output group: MSE, RMSE, MAE,  $R^2$ , AQI category accuracy, macro F1, and high-AQI recall.

# 5 System Architecture

## 5.1 Two-Repository Design

The system uses two separate GitHub repositories:

- **karAQI** (code repo): Source code, CI/CD pipelines, tests, documentation. Contains zero model files, zero data files, and zero generated artifacts.
- **karAQI-data** (data repo): Trained model files (`.joblib`), forecast JSON, evaluation metrics, weather trend images. Read by the dashboard via raw GitHub URLs.

This separation keeps the main commit history clean and ensures the data repo can be consumed by the dashboard without pulling source code.

## 5.2 Static JSON Serving

The dashboard does not run inference at runtime. Instead, CI pipelines pre-compute predictions and store them as JSON files in the `karAQI-data` repository. The Streamlit dashboard fetches these JSONs via raw GitHub URLs with a 5-minute cache, giving every visitor near-instant page load with zero runtime inference.

## 5.3 Feature Store: DuckDB

The assignment suggested Hopsworks or Vertex AI for the feature store. DuckDB was chosen as a documented alternative because:

- Hopsworks free tier experienced billing freezes and RPC disconnects on hourly feature writes.
- Dependency conflicts (`imp` module) and poor Windows support for Feast-style tooling.
- DuckDB is serverless by construction: a single file, no server, no API key, no always-on process.

The store consumes validated feature builders verbatim (no re-implementation), with a schema-versioned meta table for drift detection.

## 5.4 Model Registry: MLflow (File-Backed)

MLflow runs in file-backed mode (`models/mlruns/`): no tracking server, no paid tier, no API keys. Registered models: `aqi-hourly` (champion, hourly 30-output), `aqi-daily-h1..h3`. The forecast pipeline loads the champion model from MLflow for inference. Champion comparison ensures only models that improve on the current best are promoted; if the new model is worse, the old champion is kept.

## 5.5 CI/CD Pipelines

Three GitHub Actions workflows automate the system:

Table 4: CI/CD pipeline schedule

| Pipeline          | Schedule (UTC) | Output                                                                        |
|-------------------|----------------|-------------------------------------------------------------------------------|
| Feature pipeline  | Hourly at :01  | Incremental fetch, feature store (DuckDB)                                     |
| Forecast pipeline | Hourly at :04  | Loads champion model, reads features, fetches Open-Meteo AQ ref, exports JSON |
| Training pipeline | Daily at 00:00 | Trains Ridge + XGBoost, champion comparison, MLflow register, pushes to k     |

The forecast pipeline loads the champion model directly from the `.joblib` file in `karAQI-data`, reads features from the DuckDB feature store, runs inference, and exports the JSON. It also fetches a 96-hour (4-day) Open-Meteo AQ forecast with retry logic (3 attempts, escalating timeouts), trims it to the 72-hour window from the forecast origin, and stores it alongside the model’s 30 predictions.

**Champion comparison.** The training pipeline compares new model metrics against the current champion in MLflow. Only models that improve on the average RMSE across all output groups are promoted. If the new model is worse, the old champion is kept and the comparison is logged. This prevents model regression from bad training days.

## 5.6 Dashboard

The Streamlit dashboard presents:

- **Hero section:** Current-hour AQI (computed from observed data), live AQI from Open-Meteo, and model’s next-hour prediction.
- **72-hour forecast chart:** 30-output prediction line with Open-Meteo AQ reference line for transparency.
- **SHAP explanations:** Feature importance using LinearExplainer.
- **Model performance:** Production model metrics from the data repo, with development-only baselines as reference.
- **Weather insights:** 26-year temperature/precipitation trends and AQI seasonality analysis.

# 6 Results and Discussion

## 6.1 Hourly Models (Production)

Table 5 presents the rolling-origin evaluation for the hourly models. Two models are trained in production: Ridge and XGBoost. Development-only models (persistence, seasonal persistence, LSTM) are included for reference.

Table 5: Hourly model rolling-origin evaluation (3 expanding folds, 72h embargo)

| Model             | Group         | RMSE         | MAE          | R <sup>2</sup> | Cat. Acc.    | High-AQI Recall |
|-------------------|---------------|--------------|--------------|----------------|--------------|-----------------|
| <b>XGBoost</b>    | Hourly points | <b>8.89</b>  | <b>6.42</b>  | <b>0.845</b>   | <b>89.3%</b> | 89.3%           |
|                   | 6h means      | 18.13        | 13.12        | −0.725         | 57.8%        | 65.4%           |
|                   | 12h means     | 20.63        | 15.15        | −1.207         | 48.7%        | 44.4%           |
| Ridge             | Hourly points | 10.88        | 8.16         | 0.783          | 85.9%        | <b>85.9%</b>    |
|                   | 6h means      | <b>17.95</b> | <b>13.21</b> | <b>0.436</b>   | <b>78.1%</b> | 70.6%           |
|                   | 12h means     | <b>20.03</b> | <b>14.74</b> | <b>0.267</b>   | <b>73.5%</b> | 51.4%           |
| Persistence       | Hourly points | 14.42        | 9.63         | 0.623          | 83.7%        | 79.5%           |
|                   | 6h means      | 22.15        | 16.49        | 0.135          | 71.8%        | 63.2%           |
|                   | 12h means     | 25.37        | 19.16        | −0.164         | 66.0%        | 53.4%           |
| Seasonal persist. | Hourly points | 17.98        | 12.86        | 0.452          | 77.1%        | 71.5%           |
|                   | 6h means      | 22.93        | 16.95        | 0.079          | 70.5%        | 58.5%           |
|                   | 12h means     | 25.22        | 18.81        | −0.161         | 65.5%        | 48.0%           |
| LSTM              | Hourly points | 15.72        | 11.51        | 0.581          | 78.9%        | 72.2%           |
|                   | 6h means      | 24.97        | 19.01        | −0.092         | 66.0%        | 52.8%           |
|                   | 12h means     | 26.22        | 19.85        | −0.256         | 64.2%        | 48.5%           |

**Key findings:**

1. **XGBoost dominates on hourly points (1–24h) with RMSE 8.89, an 18% improvement over Ridge (10.88).** This matters because hourly points represent 24 of 30 outputs. XGBoost captures non-linear interactions (temperature × humidity × time-of-day) that Ridge’s linear model cannot.
2. **Ridge wins on longer ranges (25–72h) by 1–3%.** For 49–72h ahead, the signal degrades and linear models are more stable. However, XGBoost’s advantage on the primary group outweighs it in the average.
3. **Both models beat persistence on all three output groups.** This is the release gate—the models provide genuinely better forecasts than simply predicting “tomorrow = today.”
4. **Negative R<sup>2</sup> at 48–72h is expected.** The models are worse than the mean for long-range hourly predictions. However, RMSE is substantially lower than persistence (20.63 vs 25.37 for XGBoost), making the predictions useful as smoothed guidance.
5. **LSTM underperforms both Ridge and XGBoost on RMSE.** With ~35,400 training rows, the deep network has higher variance and does not outperform the simpler models.
6. **Champion comparison prevents regression.** The MLflow registry compares new model metrics against the current champion. Only models that improve on the average RMSE are promoted.

**6.2 Daily Model Comparison**

Table 6: Daily model chronological holdout results (development reference)

| Model         | +1d RMSE     | R <sup>2</sup> | +2d RMSE     | R <sup>2</sup> | +3d RMSE     | R <sup>2</sup> |
|---------------|--------------|----------------|--------------|----------------|--------------|----------------|
| XGBoost       | <b>15.77</b> | <b>0.515</b>   | <b>19.47</b> | <b>0.259</b>   | 20.57        | 0.171          |
| Random Forest | 16.26        | 0.485          | 19.52        | 0.256          | 20.73        | 0.158          |
| Ridge         | 16.61        | 0.463          | 19.92        | 0.225          | <b>20.55</b> | <b>0.173</b>   |
| SARIMA        | 26.99        | −0.420         | 38.84        | −1.948         | 60.18        | −6.095         |
| LSTM          | 27.42        | −0.465         | 26.33        | −0.355         | 25.14        | −0.238         |
| Persistence   | 18.58        | 0.328          | 22.99        | −0.032         | 25.65        | −0.289         |

The daily training pipeline is used for model comparison only. Only the hourly model is served in production. XGBoost achieves the best results for +1 and +2 day horizons. SARIMA and LSTM both have strongly negative  $R^2$  at all horizons, indicating they do not capture the AQI dynamics well with this data volume.

### 6.3 Interpretation of Metrics

An RMSE of 8.89 AQI units for hourly points means the typical prediction error is  $\pm 9$  AQI points on a 0–500 scale. This is sufficient for:

- Category-level classification (Good vs. Unhealthy for Sensitive Groups) with 89.3% accuracy
- Trend detection (AQI rising toward hazardous levels)
- 72-hour planning guidance for outdoor activities

It is **not** precise enough for exact-hour predictions. However, it consistently beats the persistence baseline, providing genuine forecasting value.

### 6.4 SHAP Feature Importance

The SHAP LinearExplainer reveals that the top contributing features are:

1. **AQI lag 1h**—the most recent AQI value is the strongest predictor of near-future AQI.
2. **PM<sub>2.5</sub> concentration**—the dominant pollutant in Karak’s AQI calculation.
3. **Temperature and humidity**—capture diurnal and seasonal patterns.
4. **Hour sin/cos**—encode the daily cycle of pollution (morning peaks, evening subsidence).

### 6.5 AQI Seasonality

Analysis of 26 years of weather data and 4 years of AQI data reveals:

- **Summer months (May–August)** show higher PM<sub>2.5</sub> due to dust storms and biomass burning.
- **Nighttime inversions** trap pollutants, leading to AQI peaks between 6–10 AM local time.
- **Monsoon precipitation (July–September)** provides natural washout, temporarily reducing AQI.
- **Dust storm season (March–June)** correlates with PM<sub>10</sub> spikes.

## 7 Conclusion

This paper presented karAQI, a complete end-to-end machine learning system for 72-hour AQI forecasting in Karak, Pakistan. The key contributions are:

1. **A practical forecasting system** for a city with no ground-air-quality monitor, using only modeled/reanalysis data from Open-Meteo.
2. **A structured model selection methodology** using weighted scoring across seven criteria, evaluating six model families and selecting XGBoost and Ridge for production.
3. **Rolling-origin evaluation with a 72-hour embargo** that prevents data leakage and provides realistic performance estimates.
4. **A serverless, zero-cost MLOps stack** with DuckDB feature store, MLflow registry, and GitHub Actions CI/CD.
5. **A static JSON serving architecture** that pre-computes predictions via CI, giving every dashboard visitor near-instant page load.
6. **A two-repository design** separating code from data artifacts for clean commit history and independent consumption.
7. **Public deployment** with real-time data updates and transparent Open-Meteo comparison.

The production XGBoost model achieves 89.3% AQI category accuracy for hourly predictions (RMSE 8.89) and consistently outperforms persistence on RMSE and MAE across all output groups. Limitations include: negative  $R^2$  at 48–72h (expected for long-range hourly forecasting), reliance on modeled data



without ground-truth validation, and the inherent uncertainty of predicting air quality 72 hours in advance.

Future work could include: integrating ground-truth monitoring data if a Karak station becomes available, adding uncertainty quantification (prediction intervals), exploring attention-based transformer architectures with larger datasets, and extending the system to additional cities in Pakistan.

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